Exhaust Gas Analyzer



Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: CO Machine /Emission check machine

What it is used for

An exhaust gas analyzer measures selected gases in vehicle exhaust, commonly carbon monoxide (CO), carbon dioxide (CO2), oxygen (O2), and hydrocarbons (HC), depending on the instrument. Correct probe placement, engine preparation, and instrument checks help assess emissions and support maintenance or compliance testing under the applicable procedure.

Main parts

- Analyzer unit and display
- Exhaust sampling probe and flexible hose
- Gas sampling filter and water trap
- Calibration or zero-check controls
- Engine speed or temperature input where fitted
- Printer, data interface, or results storage

Before you start

1. Operate the analyzer only if trained and authorised for vehicle emissions testing.
2. Confirm the vehicle fuel type, engine configuration, test procedure, and applicable emission limits.
3. Inspect the analyzer, probe, hose, filters, water trap, power supply, and connectors for damage or blockage.
4. Check calibration status, service date, and the instrument's required warm-up and self-check procedure.
5. Work in a well-ventilated area with a suitable exhaust extraction system; never run an engine in an enclosed space without adequate exhaust removal.
6. Inspect the exhaust system for major leaks and confirm that the vehicle can be tested safely under the prescribed conditions.

How to use

1. Park the vehicle in the designated test area, apply the parking brake, and secure it according to the workshop procedure.
2. Ensure the exhaust extraction system is operating and keep people away from direct exhaust discharge.
3. Switch on the analyzer and complete the manufacturer's warm-up, zero, leak, and readiness checks as applicable.

4. Select the correct vehicle and test mode, including the required measurement units and engine conditions.
5. Prepare the engine according to the approved test procedure and allow it to reach the required operating temperature where specified.
6. Insert the sampling probe into the exhaust outlet to the depth and position specified by the analyzer or test procedure.
7. Ensure the probe and hose are secure and that the sample path is not kinked, blocked, or drawing in excessive ambient air.
8. Allow the reading to stabilise and monitor the displayed gases and any instrument warnings.
9. Follow the prescribed idle or elevated-speed test sequence without exceeding the vehicle manufacturer's operating limits.
10. Record the measured values, engine conditions, vehicle identification, test mode, and any required inspection details.
11. If readings fluctuate abnormally, check for leaks, probe position, filter blockage, water-trap condition, and instrument warnings before repeating the test.
12. Remove the probe carefully after the measurement sequence and allow the analyzer to purge or recover as instructed.
13. Switch off the analyzer according to the manual, then clean or replace filters and empty the water trap using the approved procedure.
14. Report results against the applicable limits and refer failed or suspicious results for further vehicle diagnosis.

⚠ Safety

- Exhaust gases contain harmful substances, including carbon monoxide, which is colourless and potentially fatal; use adequate ventilation and exhaust extraction.
- Never perform engine-running tests in a closed or poorly ventilated area.
- Keep the sampling probe, hose, and hands away from hot exhaust components and moving engine parts.
- Use suitable eye protection and workshop PPE, and secure loose clothing around rotating components.
- Do not smoke or introduce ignition sources near fuel vapours or flammable workshop materials.
- Never exceed the manufacturer's engine-speed or test-duration limits.
- Stop the test if exhaust extraction fails, a gas alarm activates, the operator feels unwell, or unsafe engine behaviour occurs.

Care & storage

- Inspect the probe, hose, filters, water trap, and connectors after use.
- Replace filters and drain collected moisture at the intervals specified by the manufacturer.
- Perform calibration, leak checks, and gas-zero checks according to the instrument's maintenance schedule.
- Store the analyzer and probe in a clean, dry area protected from impact and contamination.

- Arrange servicing if readings drift, response becomes slow, or the instrument repeatedly fails self-checks.

Common mistakes to avoid

- Testing before the analyzer completes its required warm-up or readiness checks.
- Using a leaking or blocked sampling hose or incorrectly positioned probe.
- Ignoring ventilation and carbon monoxide exposure risks.
- Comparing readings with limits for the wrong vehicle type, fuel, or test mode.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.